Solutions to Exercises

D. Liberzon, *Calculus of Variations and Optimal Control Theory*

See the last page for the list of all exercises along with page numbers where they appear in the book.

Chapter 1

1.1

The answer is *no*.

Counterexample: on the (x_1, x_2)-plane, consider the function $f(x) = x_1(1 + x_1) + x_2(1 + x_2)$. Let D be the union of the closed first quadrant $\{(x_1, x_2) : x_1 \geq 0, x_2 \geq 0\}$ and some curve (e.g, a circular arc) directed from the origin into the third quadrant. The origin $x^* = (0, 0)$ is clearly not a local minimum, because $f(x^*) = 0$ but f is negative for small negative values of x_1 and x_2. However, it is easy to check that the listed conditions are satisfied because the feasible directions are $\{(d_1, d_2) : d_1 \geq 0, d_2 \geq 0\}$ and we have $\nabla f(x^*) = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ and $\nabla^2 f(x^*) = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$.

1.2

Example: on the (x_1, x_2)-plane, let $h_1(x) = x_1^2 - x_2$ and $h_2(x) = x_2$. Then D consists of the unique point $x^* = (0, 0)$ which is automatically a minimum of *any* function f over D. The gradients are $\nabla h_1(x^*) = \begin{pmatrix} 0 \\ -1 \end{pmatrix}$ and $\nabla h_2(x^*) = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ and they are linearly dependent, hence x^* is not a regular point. It remains to choose any function f whose gradient at x^* is not proportional to $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$—e.g., $f(x) = x_1 + x_2$ works.

See also Example 3.1.1 on pp. 279–280 in [Ber99].

Another example, a little more complicated but also more interesting, is to consider, on the (x_1, x_2)-plane, the functions $h_1(x) = x_2$ and $h_2(x) = x_2 - g(x_1)$ where

$$g(x_1) = \begin{cases} x_1^2 & \text{if } \; x_1 > 0 \\ 0 & \text{if } \; x_1 \leq 0 \end{cases}$$

Then $D = \{x : x_1 \leq 0, x_2 = 0\}$. The point $x^* = (0, 0)$ is not a regular point, and we can again easily choose f for which the necessary condition fails. The interesting thing about this example is that the tangent space to D at x^* is not even a vector space: it is a ray pointing to the left.

1.3

Let's do it for 2 constraints, then it will be obvious how to handle an arbitrary number of constraints. For $d_1, d_2, d_3 \in \mathbb{R}^n$, consider the following map from $\mathbb{R}^3$ to itself:

$$F : \begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \end{pmatrix} \mapsto \begin{pmatrix} f(x^* + \alpha_1 d_1 + \alpha_2 d_2 + \alpha_3 d_3) \\ h_1(x^* + \alpha_1 d_1 + \alpha_2 d_2 + \alpha_3 d_3) \\ h_2(x^* + \alpha_1 d_1 + \alpha_2 d_2 + \alpha_3 d_3) \end{pmatrix}.$$

The Jacobian of F at $(0,0,0)$ is

$$\begin{pmatrix} \nabla f(x^*) \cdot d_1 & \nabla f(x^*) \cdot d_2 & \nabla f(x^*) \cdot d_3 \\ \nabla h_1(x^*) \cdot d_1 & \nabla h_1(x^*) \cdot d_2 & \nabla h_1(x^*) \cdot d_3 \\ \nabla h_2(x^*) \cdot d_1 & \nabla h_2(x^*) \cdot d_2 & \nabla h_2(x^*) \cdot d_3 \end{pmatrix}.$$

Arguing exactly as in the notes, we know that this Jacobian must be singular for any choice of d_1, d_2, d_3. Since x^* is a regular point and so $\nabla h_1(x^*)$ and $\nabla h_2(x^*)$ are linearly independent, we can choose d_1 and d_2 such that the lower left 2×2 submatrix

$$\begin{pmatrix} \nabla h_1(x^*) \cdot d_1 & \nabla h_1(x^*) \cdot d_2 \\ \nabla h_2(x^*) \cdot d_1 & \nabla h_2(x^*) \cdot d_2 \end{pmatrix}$$

is nonsingular (for example, using the Gram-Schmidt orthogonalization process: choose d_1 aligned with $\nabla h_1(x^*)$ and d_2 in the plane spanned by $\nabla h_1(x^*)$ and $\nabla h_2(x^*)$ to be orthogonal to d_1). Since the Jacobian is singular, its top row must be a linear combination of the bottom two, linearly independent by construction, rows:

$$\nabla f(x^*) \cdot d_i = \lambda_1^* \nabla h_1(x^*) \cdot d_i + \lambda_2^* \nabla h_2(x^*) \cdot d_i, \qquad i = 1, 2, 3.$$

Note that the coefficients λ_1^* and λ_2^* are uniquely determined by our choice of d_1 and d_2, and do not depend on the choice of d_3. In other words, we have

$$\nabla f(x^*) \cdot d_3 = \lambda_1^* \nabla h_1(x^*) \cdot d_3 + \lambda_2^* \nabla h_2(x^*) \cdot d_3 \qquad \forall\, d_3 \in \mathbb{R}^3$$

from which it follows that $\nabla f(x^*) = \lambda_1^* \nabla h_1(x^*) + \lambda_2^* \nabla h_2(x^*)$.

1.4

This is Problem 3.1.3 in [Ber99], page 292 (an easier version appears earlier as Problem 1.1.8, page 19). The function being minimized is $f(x) = |x-y| + |x-z|$. Writing $|x-y|$ as $((x-y)^T(x-y))^{1/2}$, and similarly for $|x-z|$, it is easy to compute that

$$\nabla f(x^*) = \frac{x^* - y}{|x^* - y|} + \frac{x^* - z}{|x^* - z|}.$$

By the first-order necessary condition for constrained optimality, this vector must be aligned with the normal vector $\nabla h(x^*)$. Geometrically, the fact that the two unit vectors appearing in the above formula sum up to a constant multiple of $\nabla h(x^*)$ means that the angles they make with it are equal.

1.5, 1.6

These follow easily from the definitions of the first and second variation by writing down the Taylor expansion of $g(y(x)+\alpha\eta(x))$ around $\alpha = 0$ inside the integral:

$$J(y+\alpha\eta) = \int_0^1 g(y(x)+\alpha\eta(x))dx = \int_0^1 \big(g(y(x)) + g'(y(x))\alpha\eta(x) + \frac{1}{2}g''(y(x))\alpha^2\eta^2(x) + o(\alpha)\big)dx.$$

The second variation is

$$\delta^2 J\big|_y(\eta) = \frac{1}{2}\int_0^1 g''(y(x))\eta^2(x)dx.$$

This example also appears in Section 5.5 of [AF66].

1.7

Let $V = C^0([0,1],\mathbb{R})$ with the 0-norm $\|\cdot\|_0$, let $A = \{y \in V : y(0) = y(1) = 0, \|y\|_0 \leq 1\}$, and let $J(y) = \int_0^1 y(x)dx$. It is easy to see that A is bounded, that J is continuous, and that J does not have a global minimum over A because the infimum value of J over A is -1 but it's not achieved for any continuous curve. What's not obvious is that A is closed, because to show this we must show that if a sequence of continuous functions $\{y_k\}$ converges to some function y in 0-norm then the limit y is also continuous. The proof of this goes as follows. To show continuity of y, we must show that for every $\varepsilon > 0$ there exists a $\delta > 0$ such that when $|x_1 - x_2| < \delta$ we have $|y(x_1) - y(x_2)| < \varepsilon$. Let k be large enough so that $\|y_k - y\|_0 \leq \varepsilon/3$, and let δ be small enough so that $|y_k(x_1) - y_k(x_2)| < \varepsilon/3$ whenever $|x_1 - x_2| < \delta$ (using continuity of y_k). This gives

$$|y(x_1) - y(x_2)| \leq |y(x_1) - y_k(x_1)| + |y_k(x_1) - y_k(x_2)| + |y_k(x_2) - y(x_2)| < \varepsilon$$

and we are done. See also [Rud76, p. 150, Theorem 7.12] or [AF66, p. 103, Theorem 3-11] or [Kha02, p. 655] or [Sut75, p. 120, Theorem 8.4.1].

Chapter 2

2.1

Students submitted several interesting examples of variational problems, including:

- Choose a curve connecting two given points which minimizes the surface area obtained by rotating the curve around the x-axis. The formulation ends up being very similar to brachistochrone. Or can minimize the enclosed volume, or minimize drag during horizontal motion (the latter problem was studied by Newton).
- Among curves enclosing a given area, minimize the curve length. (Motivation: minimize the cost of building a fence.) This is "dual" to Dido's problem.
- Drive a car to a destination, or make an airplane reach desired cruising altitude, while minimizing fuel consumption.
- Consider a boat moving at constant speed across the river with fixed current. Choose a path to cross the river in minimal time.
- Find a shortest path (geodesic) connecting two points on a curved surface (such as a sphere).
- Given a source of energy (desirable or dangerous) which decays inversely proportional to square of distance from this source (or some other law), choose a curve connecting two given points which minimizes/maximizes the energy integral.

Another interesting problem (from p. 156 of the Russian book on calculus of variations by Lavrentiev and Lyusternik): determine the closed curve such that an airplane flying along this curve, with fixed wind direction and velocity, encloses maximal area in a given time.

2.2

This exercise is on page 341 in [Jur96].

$y \equiv 0$ is a weak minimum because it gives zero cost and for any other curve close enough to it in the sense of the 1-norm we have that $y'(x)$ is close to 0 for all x and the cost is nonnegative as long as $|y'(x)| \leq 1$.

On the other hand, we can construct curves that are close to 0 in the sense of the 0-norm but are "spiky" like in the example right before this exercise. For these curves $|y'|$ can be arbitrarily large and we can make the cost approach $-\infty$. So $y \equiv 0$ is a not a strong minimum.

The same construction can be done around any other curve, hence there is no strong minimum.

2.3

Let us write down the first-order Taylor expansion (mean value theorem) for $L(x, y(x)+\alpha\eta(x), y'(x)+\alpha\eta'(x))$ as a function of α around $\alpha = 0$:

$$\begin{aligned} L(x, y(x) + \alpha\eta(x), y'(x) + \alpha\eta'(x)) &= L(x, y(x), y'(x)) \\ &+ L_y(x, y(x) + \theta\eta(x), y'(x) + \theta\eta'(x))\alpha\eta(x) + L_{y'}(x, y(x) + \theta\eta(x), y'(x) + \theta\eta'(x))\alpha\eta'(x) \end{aligned}$$

where $\theta \in [0, \alpha]$.

Notation: Functions without arguments are supposed to be evaluated at $(x, y(x), y'(x))$. When a function appears with an overbar, this indicates that it is evaluated at an "intermediate" point $(x, y(x) + \theta\eta(x), y'(x) + \theta\eta'(x))$.

Also, in this problem we don't want to retain the parameter α, so we set $\alpha = 1$ in the above expression:

$$\begin{aligned} L(x, y + \eta, y' + \eta') &= L(x, y, y') + \overline{L_y}\eta + \overline{L_{y'}}\eta' \\ &= L(x, y, y') + L_y\eta + L_{y'}\eta' + (\overline{L_y} - L_y)\eta + (\overline{L_{y'}} - L_{y'})\eta' \end{aligned}$$

Integrating, we get

$$J(y + \eta) = J(y) + \int_a^b (L_y\eta + L_{y'}\eta')dx + \int_a^b \big((\overline{L_y} - L_y)\eta + (\overline{L_{y'}} - L_{y'})\eta'\big)dx$$

The first integral above matches (2.14), so it is the first variation $\delta J|_y(\eta)$. The second integral is the higher-order term that we need to investigate, let us call it Δ. We have

$$\begin{aligned} |\Delta| \le (b - a) &\max_{a \le x \le b} \big\{|L_y(x, y(x) + \theta\eta(x), y'(x) + \theta\eta'(x)) - L_y(x, y(x), y'(x))| \cdot |\eta(x)|\big\} \\ + (b - a) &\max_{a \le x \le b} \big\{|L_{y'}(x, y(x) + \theta\eta(x), y'(x) + \theta\eta'(x)) - L_{y'}(x, y(x), y'(x))| \cdot |\eta'(x)|\big\} \end{aligned}$$

If we divide this by $\|\eta\|_1 = \max_{a\le x\le b} |\eta(x)| + \max_{a\le x\le b} |\eta'(x)|$, we get

$$\begin{aligned} \frac{|\Delta|}{\|\eta\|_1} \le (b - a) &\max_{a \le x \le b} \big\{|L_y(x, y(x) + \theta\eta(x), y'(x) + \theta\eta'(x)) - L_y(x, y(x), y'(x))|\big\} \\ + (b - a) &\max_{a \le x \le b} \big\{|L_{y'}(x, y(x) + \theta\eta(x), y'(x) + \theta\eta'(x)) - L_{y'}(x, y(x), y'(x))||\big\} \end{aligned}$$

and it is clear that as $\|\eta\|_1$ goes to 0 this ratio approaches 0 because $y + \theta\eta, y' + \theta\eta'$ approach y, y' (uniformly over x) and L is $\mathcal{C}^1$ hence its partials are uniformly continuous on the finite interval $[a, b]$. This shows that indeed $\Delta = o(\|\eta\|_1)$.

The answer to the last question is *no*, because η' appears in Δ and so if we divide $|\Delta|$ by $\|\eta\|_0 = \max_{a\le x\le b} |\eta(x)|$ and let this 0-norm go to 0, the ratio is not guaranteed to be small.

2.4

This is called the DuBois-Reymond Lemma; see, e.g., pp. 63–64 in [Mac05], also p. 180 in [Lue69].

Define c to be the average value of ξ, i.e.,

$$c := \frac{\int_a^b \xi(z)dz}{b-a}$$

Define a new function $\bar{\xi}$ by

$$\bar{\xi}(x) := \xi(x) - c$$

For any η satisfying $\eta(a) = \eta(b) = 0$ we have

$$\int_a^b \bar{\xi}(x)\eta'(x)dx = \int_a^b \xi(x)\eta'(x)dx - \underbrace{c\int_a^b \eta'(x)dx}_{=c(\eta(b)-\eta(a))=0} = \int_a^b \xi(x)\eta'(x)dx = 0$$

so $\bar{\xi}$ satisfies the same condition as ξ.

Now, define

$$\eta(x) := \int_a^x \bar{\xi}(z)dz$$

For this η, we first have $\eta(a) = 0$ and

$$\eta(b) = \int_a^b \bar{\xi}(z)dz = \int_a^b \xi(z)dz - c(b-a) = 0$$

by our definition of c, so this η is admissible. Next,

$$\int_a^b \bar{\xi}(x)\eta'(x)dx = \int_a^b (\bar{\xi}(x))^2 dx$$

and this must be 0, so $\bar{\xi} \equiv 0$ which implies that $\xi \equiv c$.

2.5

This result is classical; see, e.g., [Mac05], [SW97], [You80].

The Lagrangian is $L = \frac{\sqrt{1+(y')^2}}{\sqrt{y}}$. By the "no x" result, the quantity

$$L_{y'}y' - L = \frac{(y')^2}{\sqrt{1+(y')^2}\sqrt{y}} - \frac{\sqrt{1+(y')^2}}{\sqrt{y}} = -\frac{1}{\sqrt{1+(y')^2}\sqrt{y}}$$

must be constant, so we have

$$(1+(y')^2)y = C$$

for some constant C. Make the substitution $y'(x) = \tan\alpha$ (this is valid as long as y' doesn't take the same value twice). Then $y = C/(1+\tan^2\alpha) = C\cos^2\alpha$ and so $\frac{dy}{d\alpha} = -2C\cos\alpha\sin\alpha$. Next, using a well-known trig identity,

$$\frac{dx}{d\alpha} = \frac{dx}{dy}\frac{dy}{d\alpha} = -2C\cos^2\alpha = -C(\cos 2\alpha + 1) =: -C(\cos\beta + 1)$$

In terms of $\beta := 2\alpha$ we have $\frac{dx}{d\beta} = -\frac{C}{2}(\cos\beta+1)$, which after integration gives $x = -\frac{C}{2}(\sin\beta+\beta)+D$ where D is another constant. Letting $a := D - C\pi/2$, $c := C/2$, and $\theta := \pi - \beta$, we finally arrive at the equations $x = c(\theta - \sin\theta) + a$ and $y = C\cos^2\alpha = \frac{C}{2}(1+\cos\beta) = c(1-\cos\theta)$ which match (2.7).

2.6

This is solved in [GF63, Chapter 3] (although it relies on the general formula for the variation of a functional) and in [SW77, Chapter 3].

Let $y : [a, x_f] \to \mathbb{R}$ be an optimal curve. Since the terminal point is not fixed, let the terminal point of the perturbed curve be $x_f + \alpha\Delta x$ and let the perturbed curve itself on this new interval be $y + \alpha\eta$. (Note that this is similar to our later derivation of the Weierstrass-Erdmann corner conditions.) The cost of the perturbed curve is

$$J(y + \alpha\eta) = \int_a^{x_f + \alpha\Delta x} L(x, y(x) + \alpha\eta(x), y'(x) + \alpha\eta'(x))dx$$

The first variation, which is the derivative of this with respect to α at $\alpha = 0$, is

$$\begin{aligned} \delta J\big|_y(\eta) &= \int_a^{x_f} (L_y\eta + L_{y'}\eta')dx + L(x_f, y(x_f), y'(x_f))\Delta x \\ &\underset{\text{by parts}}{=} \int_a^{x_f} (L_y\eta - \frac{d}{dx}L_{y'}\eta)dx + L_{y'}\eta\big|_a^{x_f} + L(x_f, y(x_f), y'(x_f))\Delta x \end{aligned}$$

Since perturbations η preserving the terminal point ($\Delta x = 0$) are still allowed, the usual Euler-Lagrange equation must still hold and so the integral equals 0. Also, $\eta(a) = 0$. We are left with

$$L_{y'}(x_f, y(x_f), y'(x_f))\eta(x_f) + L(x_f, y(x_f), y'(x_f))\Delta x = 0 \tag{7.39}$$

But $\eta(x_f)$ and Δx are related, because the terminal point of the perturbed curve must still be on the curve $y = \varphi(x)$:

$$y(x_f + \alpha\Delta x) + \alpha\eta(x_f + \alpha\Delta x) = \varphi(x_f + \alpha\Delta x)$$

Let's differentiate this relation with respect to α and set $\alpha = 0$:

$$y'(x_f)\Delta x + \eta(x_f) = \varphi'(x_f)\Delta x$$

Solving this for $\eta(x_f)$ and plugging the result into (7.39) gives the desired transversality condition

$$L(x_f, y(x_f), y'(x_f)) + L_{y'}(x_f, y(x_f), y'(x_f))(\varphi'(x_f) - y'(x_f)) = 0$$

(Δx is eliminated since it is arbitrary).

For the case of the length functional, the transversality condition reduces to $1 + \varphi'(x_f)y'(x_f) = 0$, which is still orthogonality (of the tangent vectors $(1, \varphi')$ and $(1, y')$ at the terminal point).

2.7

We see immediately from the second canonical equation that in the "no y" case the momentum is constant. In the "no x" case, we compute dH/dx and see that it is 0 because of the two canonical equations and because $H_{y'}$ is 0 (as noted also in (2.31) right after the exercise), hence the Hamiltonian is constant.

2.8

We just need to work with $q \in \mathbb{R}^{3N}$ (the vector of generalized coordinates). The kinetic energy is the sum over all particles, same for momentum. The Hamiltonian equals the total energy of all particles.

Interaction forces between particles are not a problem because they are absorbed into the potential function (i.e., they are conservative) as long as they depend on distance only. For more information, see Sections 10 and 13 in [Arn89].

2.9

The map is

$$F : \begin{pmatrix} \alpha_1 \\ \alpha_2 \end{pmatrix} \mapsto \begin{pmatrix} J(y + \alpha_1 \eta_1 + \alpha_2 \eta_2) \\ C(y + \alpha_1 \eta_1 + \alpha_2 \eta_2) \end{pmatrix}$$

Its Jacobian at $\binom{0}{0}$ is

$$\begin{pmatrix} \delta J|_y (\eta_1) & \delta J|_y (\eta_2) \\ \delta C|_y (\eta_1) & \delta C|_y (\eta_2) \end{pmatrix}$$

If y is not an extremal of C, then we can pick η_1 such that $\delta C|_y (\eta_1) \neq 0$. Then there exists a λ^* such that $\delta J|_y (\eta_1) + \lambda^* \, \delta C|_y (\eta_1) = 0$. Arguing exactly as in the finite-dimensional case, we can show that the Jacobian is singular. Thus $\delta J|_y (\eta_2) + \lambda^* \, \delta C|_y (\eta_2) = 0$ for arbitrary η_2, which gives the claim (with the help of integration by parts and Lemma 2.1 exactly as in our original derivation of the Euler-Lagrange equation).

2.10

These results are classical; see, e.g., [GF63], [Mac05].

Dido: $L = y$, $M = \sqrt{1+(y')^2}$. The Euler-Lagrange equation for $L + \lambda M$ gives

$$\frac{d}{dx}\frac{\lambda y'}{\sqrt{1+(y')^2}} = 1$$

hence

$$\frac{\lambda y'}{\sqrt{1+(y')^2}} = x + c$$

for some constant c. Squaring both sides and rearranging terms, we get

$$(\lambda^2 - (x+c)^2)(y')^2 = (x+c)^2$$

or

$$y' = \pm\frac{x+c}{\sqrt{\lambda^2-(x+c)^2}}$$

This can be integrated:

$$y = \mp\sqrt{\lambda^2-(x+c)^2} + d$$

where d is another constant. From this we finally get

$$(y-d)^2 + (x+c)^2 = \lambda^2$$

which is the equation of a circle.

Catenary: $L = y\sqrt{1+(y')^2}$, $M = \sqrt{1+(y')^2}$. Let us use the "no x" result:

$$(L+\lambda M)_{y'}y' - (L+\lambda M) = (y+\lambda)\Big(\frac{(y')^2}{\sqrt{1+(y')^2}} - \sqrt{1+(y')^2}\Big) = -(y+\lambda)/\sqrt{1+(y')^2}$$

must be constant, so $y + \lambda = \sqrt{1+(y')^2}\,c$ for some constant c. Squaring and solving for $(y')^2$ we have

$$(y')^2 = \frac{(y+\lambda)^2}{c^2} - 1$$

or

$$\frac{dy}{\sqrt{(y+\lambda)^2 - c^2}} = \frac{dx}{c}$$

Integrating this (using integration table) gives

$$\cosh^{-1}\Big(\frac{y+\lambda}{c}\Big) = \frac{x+d}{c}$$

where d is another constant. Therefore

$$y = c\cosh\Big(\frac{x+d}{c}\Big) - \lambda$$

which is the catenary curve (2.3) modulo parallel translation in the (x, y)-plane.

2.11

$L = T - U = \frac{1}{2}m(\dot{x}^2 + \dot{y}^2) - mgy$, $M = x^2 + y^2 - \ell^2$. Applying the Euler-Lagrange equation to $L + \lambda(t)M$ componentwise (i.e., separately for x and for y), we easily get the equations

$$\begin{aligned} m\ddot{x} &= 2\lambda(t)x \\ m\ddot{y} &= 2\lambda(t)y - mg \end{aligned}$$

It is difficult to obtain the equations of motion from these equations directly since we don't know what $\lambda(t)$ is. On the other hand, if we make the substitution $x/y = \tan\theta$ then after some calculations $\lambda(t)$ cancels out and we recover (2.55).

2.12

We use the notation introduced in the solution to Exercise 2.3. The $o(\alpha^2)$ term can be written as

$$\frac{1}{2}\int_a^b \big((\overline{L_{yy}} - L_{yy})\eta^2 + 2(\overline{L_{yy'}} - L_{yy'})\eta\eta' + (\overline{L_{y'y'}} - L_{y'y'})(\eta')^2\big)dx \cdot \alpha^2$$

We can integrate the middle term by parts to bring this expression into the desired form, with

$$\bar{P} := \frac{1}{2}(\overline{L_{y'y'}} - L_{y'y'}), \qquad \bar{Q} := \frac{1}{2}(\overline{L_{yy}} - L_{yy}) - \frac{1}{2}\frac{d}{dx}(\overline{L_{yy'}} - L_{yy'})$$

Arguing as in the solution to Exercise 2.3, we can show $\bar{P}$ and $\bar{Q}$ are of order $o(\|\alpha\eta\|_1)$, from which the claim follows, but not $o(\|\alpha\eta\|_0)$.

Alternatively, if we want to avoid integration by parts, we can go one step further in the Taylor expansion and express the $o(\alpha^2)$ term in terms of third-order partial derivatives:

$$\frac{1}{6}\int_a^b \big(\overline{L_{yyy}}\eta^3 + 3\overline{L_{yyy'}}\eta^2\eta' + 3\overline{L_{yy'y'}}\eta(\eta')^2 + \overline{L_{y'y'y'}}(\eta')^3\big)dx \cdot \alpha^3$$

We can then define

$$\bar{P} := \frac{\alpha}{2}\overline{L_{yy'y'}}\eta + \frac{\alpha}{6}\overline{L_{y'y'y'}}\eta', \qquad \bar{Q} := \frac{\alpha}{6}\overline{L_{yyy}}\eta + \frac{\alpha}{2}\overline{L_{yyy'}}\eta'$$

and these still have the required properties.

2.13

This is discussed (with some proof details omitted) in Section 27 of [GF63]; see also Exercise 9.38 in [Mac05].

The Euler-Lagrange equation for v is

$$L_y(x, y+v, y'+v') = \frac{d}{dx} L_{y'}(x, y+v, y'+v') \tag{7.40}$$

In the notation of the solution to Exercise 2.3 (after setting $\alpha = 1$ there and with v in place of η) the left-hand side of (7.40) is

$$L_y + \overline{L_{yy}}v + \overline{L_{yy'}}v' \tag{7.41}$$

As for the right-hand side of (7.40), we first write

$$L_{y'}(x, y+v, y'+v') = L_{y'} + \overline{L_{yy'}}v + \overline{L_{y'y'}}v'$$

and then, taking $\frac{d}{dx}$, we get

$$\frac{d}{dx}L_{y'} + \Big(\frac{d}{dx}\overline{L_{yy'}}\Big)v + \overline{L_{yy'}}v' + \Big(\frac{d}{dx}\overline{L_{y'y'}}\Big)v' + \overline{L_{y'y'}}v'' \tag{7.42}$$

On the other hand, the desired equation (2.68), in view of the definitions of P and Q, is

$$L_{yy}v - \Big(\frac{d}{dx}L_{yy'}\Big)v - \Big(\frac{d}{dx}L_{y'y'}\Big)v' - L_{y'y'}v'' \approx 0 \tag{7.43}$$

where $\approx$ is equality up to higher-order terms that we need to study. Writing down that (7.41) and (7.42) are equal, and also that y satisfies the Euler-Lagrange equation, we get some cancellations and indeed arrive at (7.43) where the higher-order terms on the right-hand side are

$$(L_{yy} - \overline{L_{yy}})v + \Big(\frac{d}{dx}\overline{L_{yy'}} - \frac{d}{dx}L_{yy'}\Big)v + \Big(\frac{d}{dx}\overline{L_{y'y'}} - \frac{d}{dx}L_{y'y'}\Big)v' + \Big(\overline{L_{y'y'}} - L_{y'y'}\Big)v''$$

As in the previous exercises, we see that these are of order $o(\|v\|)$, but the norm must be the 2-norm due to the presence of v'' (even though [GF63] seems to imply that this should work with the 1-norm).

2.14

With the action Lagrangian, we have $L_{\dot{q}\dot{q}} = mI_{3\times 3}$ which is positive definite. Also, if the time interval is sufficiently small, the presence of conjugate points on it can be ruled out. Thus the second-order sufficient condition applies (after a proper change of notation) and shows that extremals are automatically minima.

Chapter 3

3.1

Close to the corner point we are changing the derivative: in the interval $[c, c+\alpha\Delta x]$ between the old and the new corner point we have $\|y_\alpha - y\|_1 \approx |y_1' - y_2'|$. So, y_α is not close to y with respect to the 1-norm. But to derive the first W-E corner condition we can work with $\Delta x = 0$. Since Δy can still be arbitrary, continuity of $L_{y'}$ still follows (see the last display in the proof).

3.2

We have $L = (y')^3$, $L_{y'} = 3(y')^2$, $y'L_{y'} - L = 2(y')^3$. Since this is the "no x" case, y' must be constant, so the piecewise $\mathcal{C}^1$ extremals must be piecewise linear. Moreover, to satisfy both W-E corner conditions, the slope cannot change at the corners, so they must be straight lines. The only line satisfying the boundary conditions is $y \equiv 0$. It is neither a weak nor a strong minimum (this is easy to see by constructing perturbations). This example appears on pp. 210 and 220 in [Mac05].

3.3

The first W-E corner condition follows immediately from the integral form of the Euler-Lagrange equation: since $L_{y'}$ is an integral, it must be continuous.

The Weierstrass necessary condition was stated for non-corner points. However, by taking the limit from the left/right approaching a corner point and using continuity of $L_{y'}$, we easily see that it also holds at corner points in the sense that $E(x, y(x), y'(x^{\pm}), w) \geq 0$.

To show the second W-E corner condition at a corner point x, let us apply the above Weierstrass necessary condition at x^+ with $w := y'(x^-)$. We get $E(x, y(x), y'(x^+), y'(x^-)) \geq 0$. By continuity of $L_{y'}$, we have $L_{y'}(x, y(x), y'(x^+)) = L_{y'}(x, y(x), y'(x^-))$, and so we can rewrite the previous inequality as

$$y'(x^-)L_{y'}(x, y(x), y'(x^-)) - L(x, y(x), y'(x^-)) \geq y'(x^+)L_{y'}(x, y(x), y'(x^+)) - L(x, y(x), y'(x^+))$$

Interchanging the roles of x^+ and x^-, we arrive at the same inequality with the opposite sign. Thus it must actually be equality, which is exactly the second W-E corner condition.

Finally, to show Legendre's condition, write the second-order Taylor expansion of L:

$$L(x, y(x), w) = L(x, y(x), y'(x)) + (w - y'(x))L_{y'}(x, y(x), y'(x)) + \frac{1}{2}(w - y'(x))^2 L_{y'y'}(x, y(x), v)$$

where v is between $y'(x)$ and w. The Weierstrass necessary condition then implies that $L_{y'y'}(x, y(x), v) \geq 0$. In the limit as $w \to y'(x)$ we obtain Legendre's condition. Note that here we're only using the Weierstrass necessary condition locally, i.e., for w close to y', which confirms that it holds for weak minima as well. Also, corner points can be handled as before by working with $x^{\pm}$.

See also [Mac05], pp. 216–217.

3.4

The first set of hypotheses is: f is continuous in t and u and $\mathcal{C}^1$ in x; f_x is continuous in t and u; and $u(\cdot)$ is piecewise continuous in t. Consider $\bar{f}$ as in (3.20). It is piecewise continuous in t because (continuous)$\circ$(piecewise continuous) is piecewise continuous. We also clearly have that $\bar{f}$ is $\mathcal{C}^1$ in x and $\bar{f}_x$ is piecewise continuous in t. Therefore, existence and uniqueness of solutions is guaranteed for $\dot{x} = \bar{f}(t,x)$ and hence for the control system under every piecewise continuous control.

The second set of hypotheses is: f is continuous in t and u; $u(\cdot)$ is piecewise continuous in t; and we have, locally in (t,x,u), the Lipschitz property

$$|f(t,x_1,u) - f(t,x_2,u)| \le L|x_1 - x_2| \tag{7.44}$$

If t is bounded then, since u is piecewise continuous, $u(t)$ is also bounded. Thus we have, locally in (t,x),

$$|\bar{f}(t,x_1) - \bar{f}(t,x_2)| = |f(t,x_1,u(t)) - f(t,x_2,u(t))| \le L|x_1 - x_2|$$

by (7.44). The rest is as before.

Further relaxation: f and f_x can be only piecewise continuous in t. On the other hand, continuity of f in u cannot be relaxed to piecewise continuity, because (piecewise continuous)$\circ$(piecewise continuous) is not necessarily piecewise continuous (just think of a composite function $f \circ g$ where g is constant and equal to a value of discontinuity of f).

See [Kha02, Son98, AF66] for more information (specific places in these books are given in the Notes and References).

3.5

The Hamiltonian is $H = p_1 u_1 + \cdots + p_n u_n - L(t, x_1, \ldots, x_n, u_1, \ldots, u_n)$. The adjoint equations are $\dot{p}_i^* = L_{x_i}|_*$. We have $H_{u_i}|_* = p_i^* - L_{\dot{x}_i}|_*$ and this must be 0, hence $p_i^* = L_{\dot{x}_i}|_*$. Comparing this with the adjoint equation, we get $\frac{d}{dt} L_{\dot{x}_i}|_* = L_{x_i}|_*$ which are the Euler-Lagrange equations.

3.6

For simplicity, let's start with the case $n = 2$, $k = 1$. In this case $\dot{x}_1 = f(x_1, x_2, u)$, $\dot{x}_2 = u$, $L = L(x_1, x_2, \dot{x}_1, \dot{x}_2)$, $H = p_1 f(x_1, x_2, u) + p_2 u - L(x_1, x_2, f(x_1, x_2, u), u)$, and we must have (omitting stars)

$$0 = H_u = p_1 f_u + p_2 - L_{\dot{x}_1} f_u - L_{\dot{x}_2} \tag{7.45}$$

The adjoint equation is

$$\begin{aligned}\dot{p}_1 &= -H_{x_1} = -p_1 f_{x_1} + L_{x_1} + L_{\dot{x}_1} f_{x_1}\\ \dot{p}_2 &= -H_{x_2} = -p_1 f_{x_2} + L_{x_2} + L_{\dot{x}_1} f_{x_2}\end{aligned}$$

Define

$$\lambda^* = p_1^* - L_{\dot{x}_1}|_*$$

The augmented Lagrangian is

$$L(x_1, x_2, \dot{x}_1, \dot{x}_2) + (p_1 - L_{\dot{x}_1})(\dot{x}_1 - f(x_1, x_2, \dot{x}_2)) =: \overline{L}$$

Calculating its partial derivatives: $\overline{L}_{x_1} = L_{x_1} - (p_1 - L_{\dot{x}_1}) f_{x_1}$, $\overline{L}_{x_2} = L_{x_2} - (p_1 - L_{\dot{x}_1}) f_{x_2}$, $\overline{L}_{\dot{x}_1} = \cancel{L_{\dot{x}_1}} + p_1 - \cancel{L_{\dot{x}_1}}$, $\overline{L}_{\dot{x}_2} = L_{\dot{x}_2} - (p_1 - L_{\dot{x}_1}) f_u = p_2$ where the last equality follows from (7.45). This gives

$$\frac{d}{dt}\overline{L}_{\dot{x}_1} = \dot{p}_1 = -p_1 f_{x_1} + L_{x_1} + L_{\dot{x}_1} f_{x_1} = \overline{L}_{x_1}$$

which is the first Euler-Lagrange equation, and next,

$$\frac{d}{dt}\overline{L}_{\dot{x}_2} = \dot{p}_2 = -p_1 f_{x_2} + L_{x_2} + L_{\dot{x}_1} f_{x_2} = \overline{L}_{x_2}$$

which is the second Euler-Lagrange equation.

In the general case, the definition of the Lagrange multipliers is

$$\lambda_i^* = p_i^* - L_{\dot{x}_i}|_*, \qquad i = 1, \ldots, k$$

The calculations are similar but more tedious. They can be found in Chapter 5 of [PBGM62].

3.7

Straightforward calculation.

3.8

a) $H = p^T Ax + p^T Bu - x^T Qx - u^T Ru$. Adjoint equation is $\dot{p} = -A^T p - 2Qx$. Since $0 = H_u = B^T p - 2Ru$, the formula for the optimal control is $u^* = \frac{1}{2}R^{-1}B^T p^*$.

b) The Hessian of H is

$$\nabla^2 H = -\begin{pmatrix} 2Q & 0 \\ 0 & 2R \end{pmatrix}$$

The second variation is $\delta^2 J\big|_{u^*}(\xi) = \int_{t_0}^{t_1}(\eta^T Q\eta + \xi^T R\xi)dt$ and this is positive for each nonzero control perturbation ξ. In fact, due to the linear-quadratic structure of the problem, here the perturbed trajectory depends linearly on α: $x = x^* + \alpha\eta$ (clear, e.g., from variation of constants formula), and there are in fact no terms of order higher than α^2 in $J(u^* + \alpha\eta) - J(u^*)$. So, positivity of the second variation allows us to conclude that u^* is indeed a minimum, and in fact a global one.

See [AF66], Corollary 5-2 and Exercise 5-13, pp. 270-271, and solution on p. 763. More on this class of problems in Chapter 6.

Chapter 4

4.1

A solution can be found in [SW97].

First, note that brachistochrone is a free-time, fixed-endpoint problem. So, we should apply the maximum principle for the Basic Fixed-Endpoint Control Problem. The Hamiltonian is $H = p_1 u_1 \sqrt{y} + p_2 u_2 \sqrt{y} - p_0 \equiv 0$ along the optimal trajectory. It is maximized by choosing the control vector to be proportional to the adjoint vector and satisfying the magnitude constraint:

$$\begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = \frac{1}{p_1^2 + p_2^2} \begin{pmatrix} p_1 \\ p_2 \end{pmatrix}$$

The adjoint equation is

$$\begin{aligned} \dot{p}_1 &= 0 \\ \dot{p}_2 &= -\frac{p_1 u_1 + p_2 u_2}{2\sqrt{y}} = -\frac{|p|}{2\sqrt{y}} \end{aligned}$$

We see that p_1 is a constant. If it is 0, then u_1 is also identically 0 and we get a vertical downward motion. Otherwise, u_1 is never 0 and we can parameterize the curves[4] by x. We have $y'(x) = p_2/p_1$ and

$$y''(x) = \frac{1}{p_1}\frac{dp_2}{dx} = \frac{1}{p_1}\frac{\dot{p}_2}{\dot{x}} = -\frac{|p|^2}{2p_1^2 y}$$

From this we get the equation

$$1 + (y')^2 + 2yy'' = 0$$

which is the Euler-Lagrange equation for brachistochrone, and we already know that its solutions are cycloids.

The issue of existence of optimal curves still needs to be settled, and this can be done using the existence results in Section 4.5.

[4]In calculus of variations, we tacitly assumed that $y = y(x)$ is a well-defined (single-valued) function; here it is part of the proof, which is an advantage of the optimal control approach.

4.2

This is Exercise 5-21 in [AF66, p. 344].

4.3

Otherwise we can, starting at $y(t_3)$, apply the time-shifted version of the optimal control piece $u^*_{[t_2,t^*]}$ and get a lower cost. More formally, if $t_3 = t_2 + \tau$, then consider

$$\bar{u}(t) := u^*(t - \tau), \qquad t \in [t_3, t^* + \tau]$$

Then the corresponding trajectory, starting at $x = x^*(t_2)$ at $t = t_3$, hits S' at time $t^* + \tau$ with cost lower than $y^*(t^*)$, a contradiction. Time-independence of the dynamics and the cost is important here:

$$\int_{t_2+\tau}^{t^*+\tau} L(x,u)dt = \int_{t_2}^{t^*} L(x,u)dt$$

In particular, no other trajectory starting from $y^*(t_1) = (x^{0,*}(t_1), x^*(t_1))$ can hit S' below $y^*(t^*)$.

4.4

Apply w_1 on the interval $(b - \varepsilon(a_1 + a_2)b, b - \varepsilon a_2 b]$ and w_2 on the interval $(b - \varepsilon a_2 b, b]$. We get

$$y(b) = y^*(b) + \varepsilon\Phi_*(b, b - \varepsilon a_2 b)\nu_{b-\varepsilon a_2 b}(w_1)a_1 + \varepsilon\nu_{b_2}(w_2)a_2 + o(\varepsilon)$$

Simply by continuity (and since b is not a discontinuity of u^*) we can write $\Phi_*(b, b-\varepsilon a_2 b) = I + O(\varepsilon)$ and $\nu_{b-\varepsilon a_2 b}(w_1) = \nu_b(w_1) + O(\varepsilon)$. Hence

$$y(b) = y^*(b) + \varepsilon\nu_b(w_1)a_1 + \varepsilon\nu_{b_2}(w_2)a_2 + o(\varepsilon)$$

and we're done. See [PBGM62].

4.5

The warping map—let's call it F—is continuous because the terminal points depend continuously on the perturbation parameters which parameterize the ball B_ε; see [PBGM62].

For any $\alpha \in (0, \varepsilon)$, the $o(\varepsilon)$ terms satisfy $|o(\varepsilon)| < \alpha\varepsilon$ if ε is small enough. For an arbitrary z in the $(1-\alpha)\varepsilon$ ball, we want to find a y in B_ε such that $F(y) = z$, which is equivalent to $y = y - F(y) + z$. The map $G(y) := y - F(y) + z$ maps B_ε to itself, hence it has a fixed point. See [Sus00], Lemma 5.3.1 ($r \leftrightarrow \varepsilon$, $\rho \leftrightarrow o(\varepsilon)$), also [LM67].

4.6

For $t' > t$,

$$
\begin{aligned}
&\lim_{t'\to t} \frac{m(x^*(t'), p^*(t')) - m(x^*(t), p^*(t))}{t'-t} \\
&\ge \lim_{t'\to t} \frac{H(x^*(t'), p^*(t'), u^*(t)) - H(x^*(t), p^*(t), u^*(t))}{t'-t} \qquad \text{(now the arguments of } u^* \text{ are the same!)} \\
&= \langle H_x|_*, \dot{x}^*(t)\rangle + \left\langle H_p|_*, \dot{p}^*(t)\right\rangle = \left\langle H_x|_*, H_p|_*\right\rangle + \left\langle H_p|_*, -H_x|_*\right\rangle = 0
\end{aligned}
$$

If we use $u^*(t')$ instead of $u^*(t)$, can check that the inequality is flipped: $\frac{dm}{dt}\big|_t \le 0$. We know that m must be differentiable as a function of time $\Rightarrow$ the time derivative is 0.

4.7

$$\dot{p}^* = -(f_x)^T\big|_* \bar{p}^* + K_{xx}|_* \, f|_* + (f_x)^T\big|_* \, K_x|_* - K_{xx}|_* \, f|_* = -(f_x)^T\big|_* p^* = -H_x|_*$$

For the second question, the reason is the same as the one behind (4.33): the above is a homogeneous LTV system, and the terminal condition is nonzero (we assumed this).

4.8

We can eliminate the terminal cost by adding $\langle K_x, f\rangle$ to the running cost. In the new problem, the Hamiltonian will be $\overline{H} = \langle \bar{p}, f\rangle + p_0(L + \langle K_x, f\rangle) + p_{n+1} = \langle \bar{p} + p_0 K_x, f\rangle + p_0 L + p_{n+1}$. The transversality condition for this new problem is $\langle \bar{p}(t_f), d\rangle = 0$ for all $d \in T_{x_f} S_1$. Now we can go back to the original problem by defining $p(t) := \bar{p}(t) + p_0 K_x(x(t))$, which gives the transversality condition

$$\langle p^*(t_f) - p_0^* K_x(x_f), d\rangle = 0 \quad \forall\, d \in T_{x_f} S_1$$

where $p_0^* \leq 0$. Canonical equations and the Hamiltonian maximization condition are the same as before. For the Hamiltonian we have $H|_*(t) = -\int_t^{t_f} H_t|_*(s)ds$ and the boundary condition $H|_*(t_f) = 0$.

Theorem 5-11 in [AF66] is close but not exactly the same, and doesn't have full details.

4.9

Perturbations of the initial points lead to infinitesimal directions of the form $(0, d_1)$ in y-space, which when propagated up to time t^* end up in the terminal cone, so we have (propagating back to initial time) $\langle p^*(t_0), d_1\rangle \leq 0$. On the other hand, at the final time we have as before $\langle p^*(t^*), d_2\rangle \geq 0$, or $\langle -p^*(t^*), d_2\rangle \leq 0$. Here, the vector $d = (d_1, d_2)$ is tangent to S_2. Combining, and using the fact that $-d$ is also in the tangent space, we get (4.46). Note that we cannot pass from inequality to equality before combining the two inequalities into one, since d_1 and d_2 may be related; e.g., $(d_1, -d_2)$ might not be in the tangent space to S_2.

4.10

The difference with the example in the text is that now we'll have the transversality condition $p_2^*(t_f) = 0$. Since p_2^* is a nonzero linear function of time, this means that $p_2^* \neq 0$ for $t < t_f$. Therefore, the optimal control is constant (doesn't switch). For $x_0 < 0$ it is $u^* \equiv 1$ and for $x_0 > 0$ it is $u^* \equiv -1$. In terms of car motions, it means apply maximal acceleration/braking to turn around if necessary and slam into the wall at 0.

This is [Sus00], "hard landing", Handout 7, p. 5.

4.11

The adjoint equation is $\dot{p}_1 = p_2$, $\dot{p}_2 = -p_1$. So, $u^*(t) = \operatorname{sgn}(c_1 \sin t + c_2 \cos t)$ for some constants c_1, c_2. We see that if we normalize by $\sqrt{c_1^2 + c_2^2}$ and use the identity $\sin^2 t + \cos^2 t = 1$, we can simplify the expression to $u^*(t) = \operatorname{sgn}(\sin(t + d))$ where $d = \arctan(c_2/c_1)$. So the control is bang-bang, and the switches occur π seconds apart. When $u = 1$, we have $\dot{x}_1 = x_2$, $\dot{x}_2 = -x_1 + 1$ and the trajectories are circles centered at $(1, 0)$. When $u = -1$, we have $\dot{x}_1 = x_2$, $\dot{x}_2 = -x_1 - 1$ and we get circles centered at $(-1, 0)$. For further details, see [PBGM62, §5] or [Kno81, p. 22].

4.12

We have $\langle p^*(t^*), e^{A(t^*-t)}Bu^*(t)\rangle = \max_{u\in U}\langle p^*(t^*), e^{A(t^*-t)}Bu(t)\rangle$. By controllability, the vector $\nu^*(t) := \langle p^*(t^*), e^{A(t^*-t)}B\rangle$ is not identically 0. The unit ball is strictly convex, so u^* is unique for almost all t and is on the boundary of the ball. In fact, it's given by the formula $u^*(t) = \nu^*(t)/|\nu^*(t)|$.

See Section 10.3 in [Son98].

4.13

Answer: $= \langle p^*, [f, g_i]\rangle|_* + \sum_{j\neq i} \langle p^*, [g_j, g_i]\rangle|_* \, u_j^*$.

4.14

Repeating the calculation in the text, we arrive at the condition that $[g, (\mathrm{ad} f)^m(g)]$ should be a linear combination of $(\mathrm{ad} f)^i(g)$, $i = 0, \ldots, m+1$:

$$[g, (\mathrm{ad} f)^m(g)](x) = \sum_{i=0}^{m+1} \alpha_i(x)(\mathrm{ad} f)^i(g)(x)$$

such that $|\alpha_{m+1}(x)| < 1$ for all x.

See [Sus79, Sus83] for details.

4.15

The following example is taken from [Sus83]. Consider the time-optimal problem in $\mathbb{R}^3$:

$$\begin{aligned}\dot{x}_1 &= x_2\\ \dot{x}_2 &= u\\ \dot{x}_3 &= x_1^2\end{aligned}$$

$|u| \leq 1$. Goal: $(x_1(0), x_2(0), -J^*) \mapsto (0,0,0)$, where J^* is the optimal cost in Fuller's problem. Then, we have exactly one trajectory that meets the boundary conditions $\Rightarrow$ it's automatically time-optimal, and it has infinitely many switches. So, the property stated earlier for time-optimal controls in $\mathbb{R}^2$ does not extend to higher dimensions.

4.16

This is Example 3.5 in [BP07]. We can reach points arbitrarily close to $(0,0)$, of the form $(0,\varepsilon)$, but $(0,0) \notin R^t$ because $\varepsilon > 0$ always (x_1 cannot remain at 0). The problem of minimizing $x_2(t_f)$ has no solution. For the convex case, we get a larger reachable set which will now be closed by Filippov's theorem. In fact, this larger reachable set will be exactly the closure of the original one, i.e., the original one is dense in the new one. This is a special instance of the so-called *relaxation theorem*. Note, however, that for *linear* systems the reachable set is closed without convexification of U—see, e.g., [BP07, p. 67].

Chapter 5

5.1

Solution from [YZ99]:

$$\begin{aligned} V(t,x) &\le J(t,x,u) \qquad (u \text{ is arbitrary}) \\ &= \int_t^{t+\Delta t} L ds + J(t+\Delta t, x(t+\Delta t), u) \end{aligned}$$

Take inf over u on the right-hand side, noting that $\inf_{u_{[t,t_1]}} = \inf_{u_{[t,t+\Delta t]}} \inf_{u_{[t+\Delta t,t_1]}}$ and the second infimum can be moved inside because the integral doesn't depend on it. This gives $V(t,x) \le \overline{V}$.

Solution from [BP07]: For arbitrary $\varepsilon > 0$, there exists $u_{[t,t+\Delta t]}$ such that

$$\int_t^{t+\Delta t} L ds + V(t+\Delta t, x(t+\Delta t)) \le \overline{V} + \varepsilon$$

and there exists $u_{[t+\Delta t,t_1]}$ such that

$$J(t+\Delta t, x(t+\Delta t), u_{[t+\Delta t,t_1]}) \le V(t+\Delta t, x(t+\Delta t)) + \varepsilon$$

Taking u to be the concatenation of these two controls, we get $J(t,x,u) \le \overline{V} + 2\varepsilon$, hence $V = \inf J$ must be $\le \overline{V}$.

5.2

The value function V will not depend on t, which simplifies the HJB equation (left-hand side becomes 0). Solving HJB directly is hard. However, we can find V from the solution based on the maximum principle in Section 4.4.1, by calculating the time it takes to reach the origin along the optimal trajectory from an arbitrary initial condition. In fact, since $|\dot{x}_2| = |u| = 1$, this time equals the total vertical distance traveled along the trajectory. This calculation is done in Example 7.3 in [BP07], p. 146. The formula for the value function is

$$V(x_1, x_2) = \begin{cases} x_2 + 2\sqrt{x_1 + \frac{1}{2}x_2^2} & \text{if } \ x_1 \geq -\frac{1}{2}|x_2|x_2 \\ -x_2 + 2\sqrt{-x_1 + \frac{1}{2}x_2^2} & \text{if } \ x_1 < -\frac{1}{2}|x_2|x_2 \end{cases}$$

It can be directly verified that this V solves the HJB equation.

5.3

This is a simple manipulation of things already mentioned. Since $\widehat{V}$ is the cost for the given control, it must satisfy the HJB equation with $\hat{u}$ plugged in, as in the second line of (5.13). Since it also satisfies the HJB equation with the inf, we conclude that $\hat{u}$ satisfies the H-maximization condition. Now we can invoke the sufficient condition. Alternatively, we can just follow the proof of the sufficient condition (the first part is not needed, just the second part).

5.4

Sufficiency consists of three conditions. First, the infinite-horizon version of the HJB equation:

$$0 = \inf_{u \in U} \left\{ L(t,x,u) + \left\langle \widehat{V}_x(t,x), f(t,x,u) \right\rangle \right\}$$

(no boundary condition). Second, the minimization condition

$$L(t,\hat{x}(t),\hat{u}(t)) + \left\langle \widehat{V}_x(t,\hat{x}(t)), f(t,\hat{x}(t),\hat{u}(t)) \right\rangle = \min_{u \in U} \left\{ L(t,\hat{x}(t),u) + \left\langle \widehat{V}_x(t,\hat{x}(t)), f(t,\hat{x}(t),u) \right\rangle \right\}$$

And third, we must assume that $\widehat{V}(x(t)) \to 0$ as $t \to \infty$ along every trajectory $x(\cdot)$ for which the infinite-horizon cost $\int_{t_0}^{\infty} L dt$ is bounded. This last condition is unpleasant (it's not easy to check) but without it the proof doesn't work. With this condition, however, the proof works almost exactly as before. Alternatively, can assume that $\widehat{V}(0) = 0$ and all bounded-cost trajectories converge to 0 (although this is stronger and not any easier to check).

5.5

$$
\begin{aligned}
\dot{p}^* &= -\frac{d}{dt}\left(V_x|_*\right) = -V_{tx}|_* - \underbrace{V_{xx}|_*}_{\text{matrix}} \cdot \dot{x}^* \\
&= \text{(swapping partials)} \ \ -V_{xt}|_* - V_{xx}|_* \cdot f|_* \\
&= -\frac{\partial}{\partial x}\left(\underbrace{V_t + \langle V_x, f\rangle}_{=-L \text{ by HJB}}\right)\Bigg|_* + (f_x)^T \underbrace{V_x|_*}_{=-p^*} \\
&= L_x|_* - (f_x)^T\Big|_* p^*
\end{aligned}
$$

and this is the correct adjoint equation.

5.6

A function with infinite slope (non-Lipschitz) like $x^{1/3}$, or a function like $v(x) = x\sin(1/x)$, both considered at $x = 0$. Or a function of two variables with a saddle point.

5.7

This is proved in [BP07]. Fix an arbitrary pair (t_0, x_0). We need to show that for every $\mathcal{C}^1$ test function $\varphi = \varphi(t, x)$ such that $\varphi - V$ attains a local maximum at (t_0, x_0) we have the inequality

$$\varphi_t(t_0, x_0) + \inf_{u \in U}\{L(t_0, x_0, u) + \langle \varphi_x(t_0, x_0), f(t_0, x_0, u)\rangle\} \le 0.$$

The negation of this statement is that there exist a $\mathcal{C}^1$ function φ satisfying

$$\varphi(t_0, x_0) = V(t_0, x_0), \qquad \varphi(t, x) \le V(t, x) \quad \forall\, (t, x) \text{ near } (t_0, x_0)$$

such that for *all* controls u we have

$$\varphi_t(t_0, x_0) + L(t_0, x_0, u) + \langle \varphi_x(t_0, x_0), f(t_0, x_0, u)\rangle > \Theta > 0.$$

Taking (t_0, x_0) as the initial condition, let us consider an arbitrary control $u(\cdot)$ and the corresponding state trajectory $x(\cdot)$ on a small interval $[t_0, t_0 + \Delta t]$. We will now see that the rate at which the value function decreases during this time interval is too slow to be consistent with the principle of optimality. As long as we pick Δt to be sufficiently small, we have

$$V(t_0 + \Delta t, x(t_0 + \Delta t)) - V(t_0, x_0) \ge \varphi(t_0 + \Delta t, x(t_0 + \Delta t)) - \varphi(t_0, x_0) = \int_{t_0}^{t_0+\Delta t} \frac{d}{dt}\varphi(t, x(t))dt$$
$$= \int_{t_0}^{t_0+\Delta t} \big(\varphi_t(t, x(t)) + \langle \varphi_x(t, x(t)), f(t, x(t), u(t))\rangle\big)dt > -\int_{t_0}^{t_0+\Delta t} L(t, x(t), u(t))dt + \Theta\Delta t$$

hence

$$V(t_0, x_0) < \int_{t_0}^{t_0+\Delta t} L(t, x(t), u(t))dt + V(t_0 + \Delta t, x(t_0 + \Delta t)) - \Theta\Delta t.$$

Taking infimum over u we get

$$V(t_0, x_0) \le \inf_{u_{[t_0, t_0+\Delta t]}} \left\{\int_{t_0}^{t_0+\Delta t} L(t, x(t), u(t))dt + V(t_0 + \Delta t, x(t_0 + \Delta t))\right\} - \Theta\Delta t.$$

On the other hand, the principle of optimality (5.4) tells us that

$$V(t_0, x_0) = \inf_{u_{[t_0, t_0+\Delta t]}} \left\{\int_{t_0}^{t_0+\Delta t} L(t, x(t), u_0)dt + V(t_0 + \Delta t, x(t_0 + \Delta t))\right\}$$

and we arrive at a contradiction.

Note that the interval $[t_0, t_0 + \Delta t]$ has to be small enough so that on this interval $\varphi_t(t, x(t)) + L(t, x(t), u(t)) + \langle \varphi_x(t, x(t)), f(t, x(t), u(t))\rangle$ remains larger than Θ for all controls and their corresponding trajectories starting at (t_0, x_0). For this, in addition to continuity, we need to assume, e.g., that U is a compact set—otherwise such a Δt may not exist. This is one place where the technical assumptions mentioned in the statement of the main result are needed (they are not needed for the part shown in the text, though, because there the control was fixed).

5.8

The sufficient condition is almost the same as in Exercise 5.3, except that we now say that $\widehat{V}$ satisfies the HJB equation in the viscosity sense. We also need to assume that f, L, and K have the properties necessary for the main result of Section 5.3.3 to hold. To prove it, just use the uniqueness claim of the main result and the definition of $\widehat{V}$ as cost. Also note that, unlike in Exercise 5.3, we can now make the claim that $\widehat{V}$ is the value function (not just that $\widehat{V}(t_0, x_0)$ is the optimal cost).

Chapter 6

6.1

To differentiate P given by (6.15) it is useful to know the time derivative of X^{-1}. From the identity

$$0 = \frac{d}{dt}(I) = \frac{d}{dt}(XX^{-1}) = X\frac{d}{dt}(X^{-1}) + \dot{X}X^{-1}$$

we get

$$\frac{d}{dt}(X^{-1}) = -X^{-1}\dot{X}X^{-1}$$

The rest is a straightforward calculation.

6.2

a) Use the fact that M is symmetric, that P^T satisfies the same differential equation with the same (symmetric) terminal condition, and uniqueness of solutions for ODEs.

b) This follows immediately from the fact that we just proved that $x^T P x$ is the optimal cost-to-go, since Q and R are positive (semi)definite.

c) $M > 0$ will do, or can assume $Q(t) > 0$ for all t, or observability (as discussed in Section 6.2.4).

6.3

The cost-to-go from time t for control u is

$$
\begin{aligned}
&\int_t^{t_1} \big(x^T Q x + u^T R u\big) dt + x^T(t_1) M x(t_1) \\
&= \int_t^{t_1} \big(x^T Q x + u^T R u\big) dt + x^T(t_1) P(t_1) x(t_1) \\
&= \int_t^{t_1} \big(x^T Q x + u^T R u\big) dt + x^T(t_1) P(t_1) x(t_1) - x^T(t) P(t) x(t) + x^T(t) P(t) x(t) \\
&= \int_t^{t_1} \Big(x^T Q x + u^T R u + \frac{d}{dt} x^T P x\Big) dt + x^T(t) P(t) x(t) \\
&\underset{\dot{x}=Ax+Bu}{=} \int_t^{t_1} \big(x^T Q x + u^T R u + x^T(\dot{P} + PA + A^T P)x + x^T P B u + u^T B^T P x\big) dt + x^T(t) P(t) x(t) \\
&\underset{\text{RDE}}{=} \int_t^{t_1} \big(x^T P B R^{-1} B^T P x + u^T R u + x^T P B u + u^T B^T P x\big) dt + x^T(t) P(t) x(t) \\
&= \int_t^{t_1} (B^T P x + R u)^T R^{-1} (B^T P x + R u) dt + x^T(t) P(t) x(t)
\end{aligned}
$$

and now the optimal control and optimal cost-to-go are obvious.

6.4

The solution of the ARE is $P = \begin{pmatrix} \pm\sqrt{3} & 1 \\ 1 & \pm\sqrt{3} \end{pmatrix}$, need to pick the one with the plus signs to get a positive semidefinite matrix. It is not very hard to solve the RDE either by using the Hamiltonian matrix as in Exercise 6.1 or just by computer simulation. The answer to the last question is *no* (as long as $M \geq 0$ of course); even though in our analysis so far we did use the fact that $M = 0$ (see, e.g., the monotonicity argument) the independence of the limit from M follows from the proof of uniqueness of the solution of the ARE in Theorem 6.1.

6.5

Controllability was used to show the cost is bounded, but stabilizability gives that too. Indeed, picking a control that makes the closed-loop system exponentially stable, we get a bounded cost. Observability was used twice: First, when proving closed-loop stability, we needed the property $y \to 0 \Rightarrow x \to 0$—detectability gives that too. Second, in statement 1, to prove $P > 0$ we needed the property $y \equiv 0 \Rightarrow x \equiv 0$—detectability doesn't give this, so we can only get $P \geq 0$. Uniqueness remains valid for $P \geq 0$ as we said in our uniqueness proof. See [KS72] for details.

6.6

It's straightforward to derive that the closed-loop system is

$$\dot{x}^* = -\sqrt{a^2 + \frac{b^2 q}{r}}\,x^*$$

from which all claims follow.

List of exercises